



National University

of Computer & Emerging Sciences Peshawar Campus



Name: Arsam Ali

Roll No: 23p0038

Program: BCS(D)/SE/AI

Semester: Fall – 2023

Time Allowed: 3hour

Course: NS1001 Applied Physics

Examination: Final

Total Marks: 50 Weightage: 50

Date: Dec 30, 2023

Instructor: Engr M Asif Khan

NOTE: Attempt all questions

Question 1

- a. A jet airliner, moving initially at 300mi/h to the east, suddenly enters a region where the wind is blowing at 100mi/h toward the direction 30° north of east. Calculate the new speed and direction of the aircraft relative to the ground. $\checkmark$ 349.41 $\times$ 2.36 [5M]
- b. A ball is tossed from an upper story window of a building. The ball is given an initial velocity of 8m/s at an angle of 20° below the horizontal. It strikes the ground 3s latter. Calculate: i) how far horizontally from the base of the building does the ball strike the ground. ii) the height from which the ball was thrown. iii) the time it takes the ball to reach a point 10m below the level of launching. $\checkmark$ 20 52.35 1.13 [5M]

Question 2

- a. For a mass attached to a spring moving horizontally (frictionless surface) and performing SHM derive the values for phase constant for the possible four cases along its verification with velocity expressions where it is necessary. $\times$ [5M]
- b. A piston in a gasoline engine is in simple harmonic motion. Taking the extremes of its position relative to its center point as ± 5 cm, calculate the maximum velocity and acceleration of the piston when the engine is running at 60Hz. $\checkmark$ 1884.91 710611.5 [5M]

Question 3

- a. Two waves $X = A \sin(kx - \omega t)$ and $Y = A \sin(kx - \omega t + \phi)$ are superposition. Derive mathematically the resultant wave function "Z". Moreover, point out the conditions for constructive and destructive interference. $\checkmark$ [10M]
- b. Two waves are travelling in the same direction along a stretched string. The waves are 90° out of phase. Each wave has amplitude of 4cm. Calculate the amplitude of the resultant wave. $\checkmark$ [10M]

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Question 4

- a. Keeping in mind the water tank analogy in what condition the current will be zero and infinite. Show your arguments with mathematical support. [5M]
- b. What is the difference between electric field and magnetic field (not more than five-sentences)? Just mention different classes (in hierarchical form) of EM spectrum. [5M]

The End